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Respondent Replay

This Help Center article explains how to use the Replay editor

The Replay editor is the place in the software where you review the data that has been recorded (or post-imported) into the iMotions software. The editor allows you to navigate through the data from sensors and/or R-Notebooks, to replay the recordings from respondent, environment, scene or screen cameras, and to [annotate](#) your data. The Replay editor is further the place where you generate scan-paths (see also: [Scan-paths](#)).

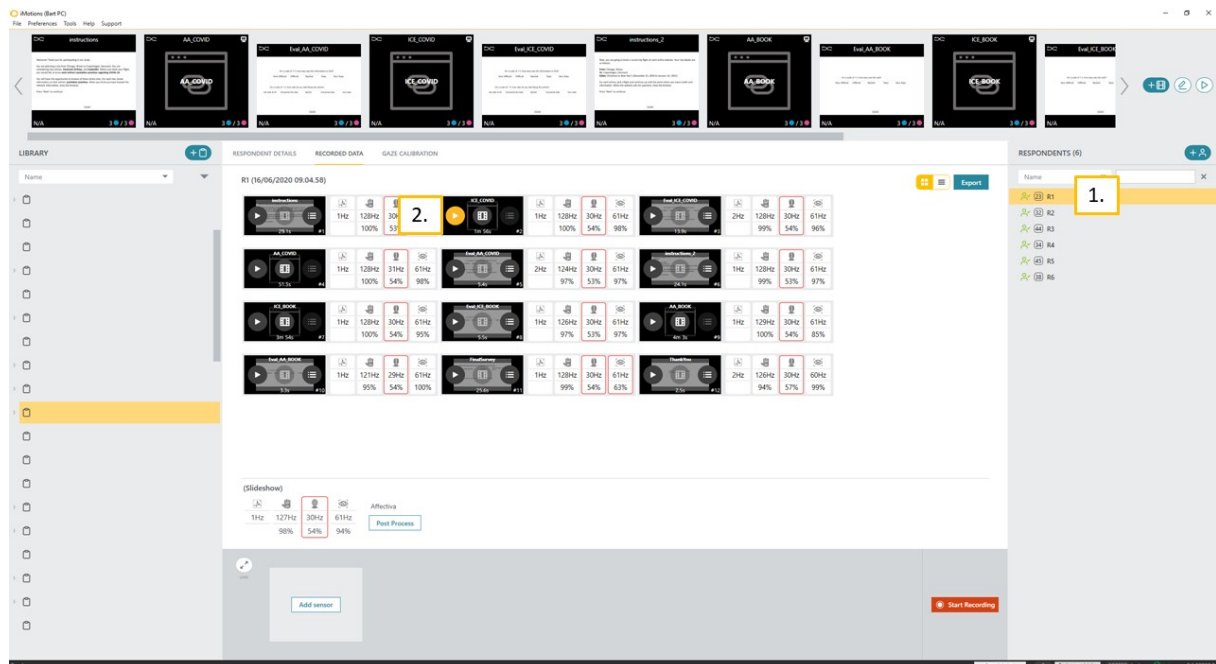
In addition to the Replay editor, there is also the Aggregate replay editor. Note that the Replay editor replays the data from *individual* respondents (consistent with how the data was originally recorded), whereas the Aggregate replay editor replays the *aggregate* data from all respondents included in the analysis (see also: [Aggregate replay](#)).

How to access the Replay editor

The Replay editor can already be accessed from the 'Study level', hence, immediately after a data collection, or from the 'Analysis level'. This is the part of the software you enter by creating an analysis, or by clicking the name of an existing analysis in the Library.

From Study Level

If you enter the Replay editor from the 'Study level' of the iMotions software (as described in this Help Center article), in the editor you (only) have access to the 'raw' data (i.e., the unprocessed data as delivered directly by the hardware systems). Hence, the primary reason for using the Replay editor from the Study level would be for assessing the data quality in preparation of the analysis.



How to access the Replay editor from the 'Study level'

Access the Replay editor from the 'Study level' (yellow path):

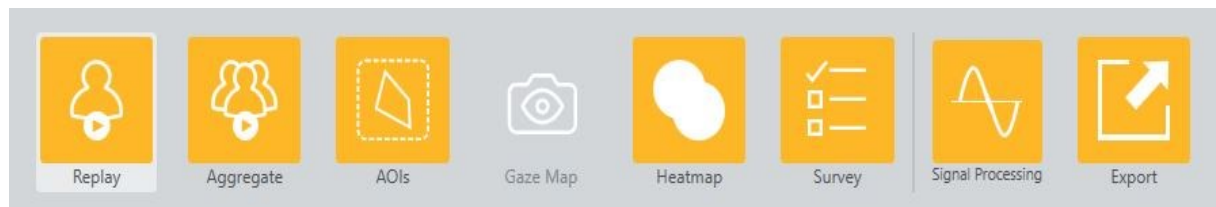
1. Select the study in the Library
2. Select a respondent from the Respondents list
3. Navigate to the 'Recorded data' page in the Respondent menu
4. Click the 'Play' button for a given stimulus

The primary reason for entering the Replay editor from the Study level (and not from the Analysis level) would be for assessing the data quality in preparation of the analysis. However, it is not unlikely that your data is not yet complete at this point. For instance, if you have collected Facial Expression Analysis data, you might still need to post-process the data first (as maximally 50% of the video frames were analyzed live). Also the data generated by R-Notebooks is not yet available (see also: [R-Notebooks](#)), including data aggregations (as these aggregations are computed by R-Notebooks) (see also: [Aggregate replay](#)).

Indeed, at this point in time only the 'raw' sensor data is available; the data as *delivered directly by the hardware systems* used to record data.

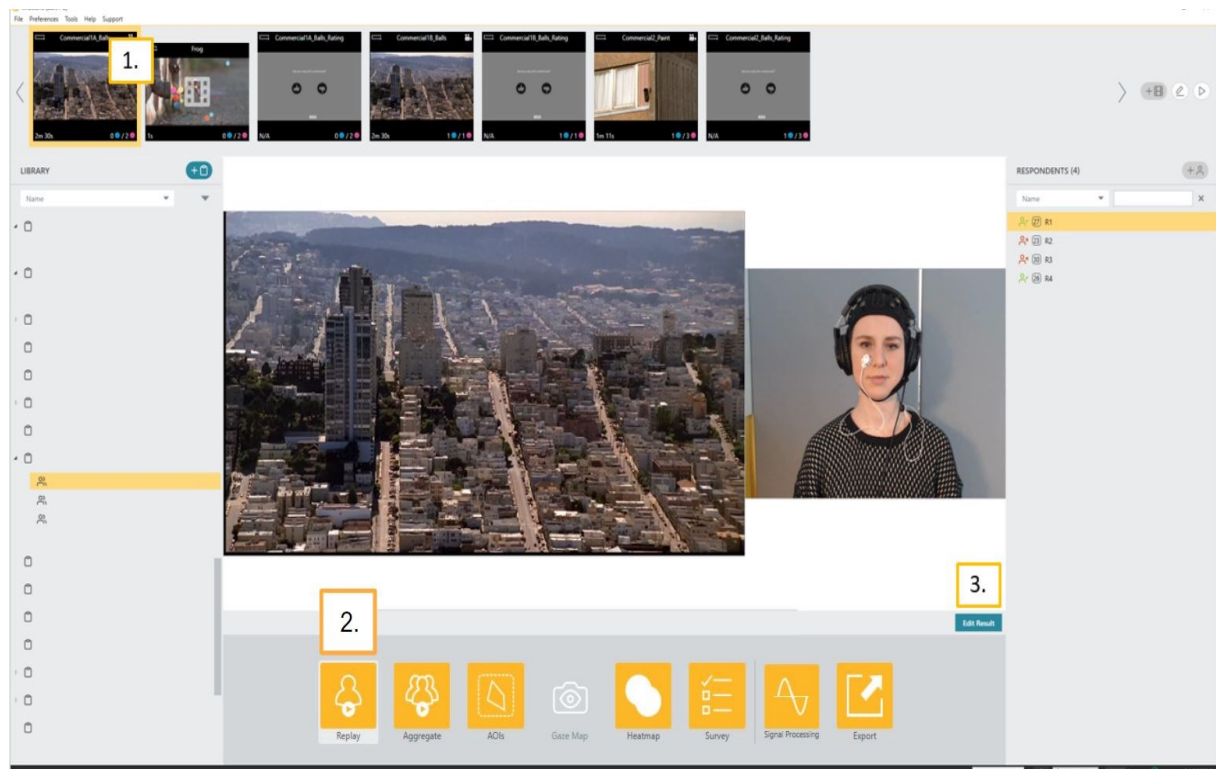
From Analysis Level

Alternatively, the Replay editor can also be accessed from the 'Analysis level' of the iMotions software. This Replay editor is nearly identical to the Replay editor from the Study level, except that more types of data are available (i.e., processed data).



Analysis menu ('Analysis level')

From here, you would access the Replay editor by choosing the Replay option in the Analysis menu.



How to access the Replay editor from the 'Analysis level'

Access the Replay editor from the 'Analysis level' (yellow path):

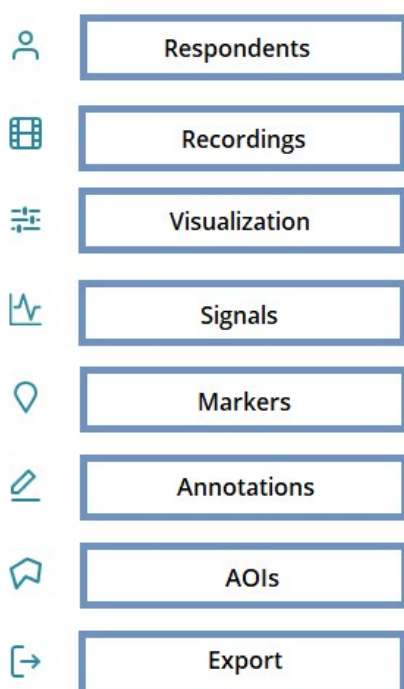
1. Click on a stimulus in the Stimulus bar
2. Click on the option Replay in the Analysis bar
3. Click Edit result

If you access the Replay editor from the Analysis level, this is typically with a different purpose than if you would have accessed the Replay editor from the Study level. At this point in the research process, the data collection has usually been completed, the necessary post-processing steps have been taken, and you may already have applied R Notebooks for signal processing the sensor data (see also: [R-Notebooks](#)). For instance, it could be that you applied a peak detection algorithm to isolate the arousal peaks in the electrodermal activity data (i.e. a measurement of emotional arousal). These additional types of data are now also available in the Replay editor.

You may now use the Replay editor to replay the (processed) sensor data, to annotate the recordings (see also: [Annotations](#)), and/or to generate Scan-paths (see also: [Scan-paths](#)).

Replay menu

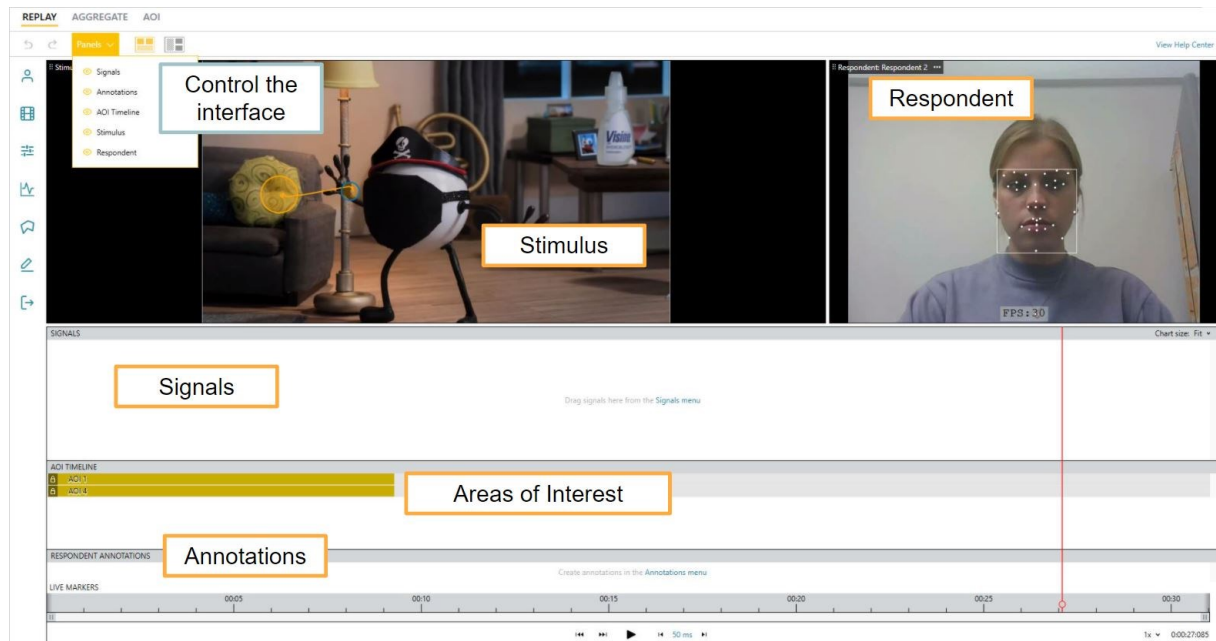
In the top-left corner of the Replay editor you find a menu, which allow you to do these things. The menu can be expanded by clicking on the icons.



Interface options

The interface options menu bar allows you to customize the Replay editor, by (un)tickling Panels, and by choosing a desired layout. For instance, you might want to enable the 'Stimulus', 'Respondent', or 'Environment' panels for replaying screen recordings, website recordings, and other media recordings, the 'Signals' panel for replaying sensor data, or the 'Annotations panel for annotating the recordings.

Moreover, if you have a setup with multiple screens, you could further choose to display the media panels in a separate window (such that you could move these panels to a secondary screen).

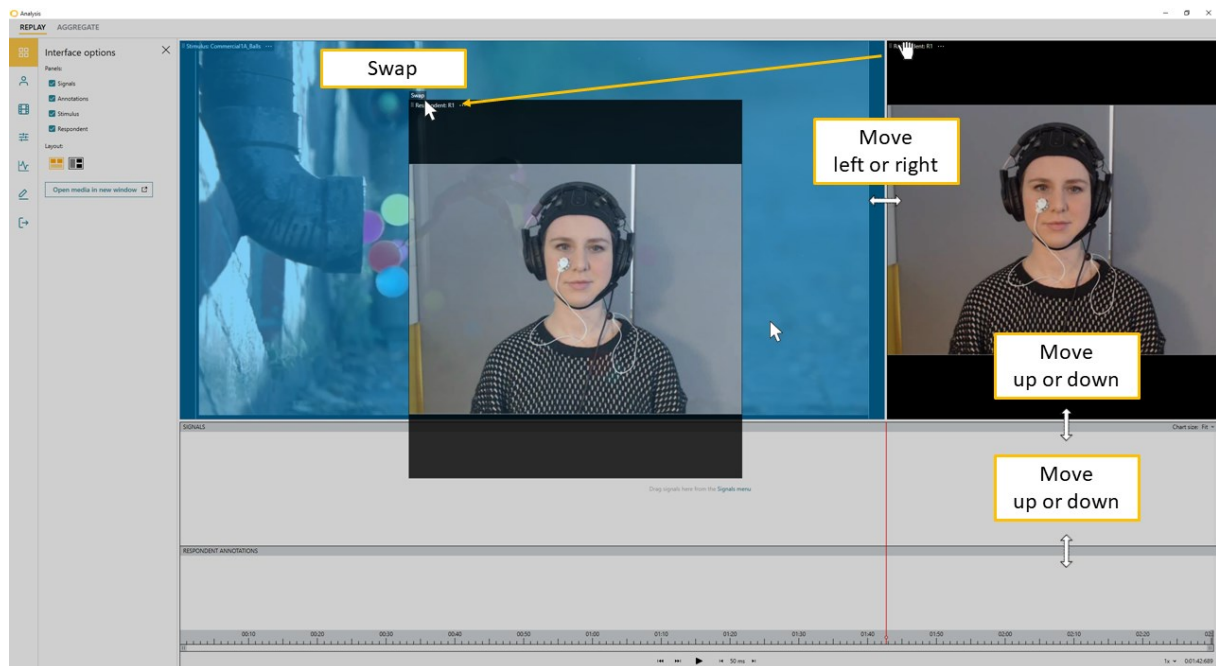


How to control the interface

Control the interface:

1. Navigate to the Interface options menu
2. (Un)tick the panels of interest
3. Choose a layout

For further customization of the layout, it is possible to enlarge the panels by dragging the borders, or by swapping the position of two panels.



How to swap and enlarge panels

Swap two panels:

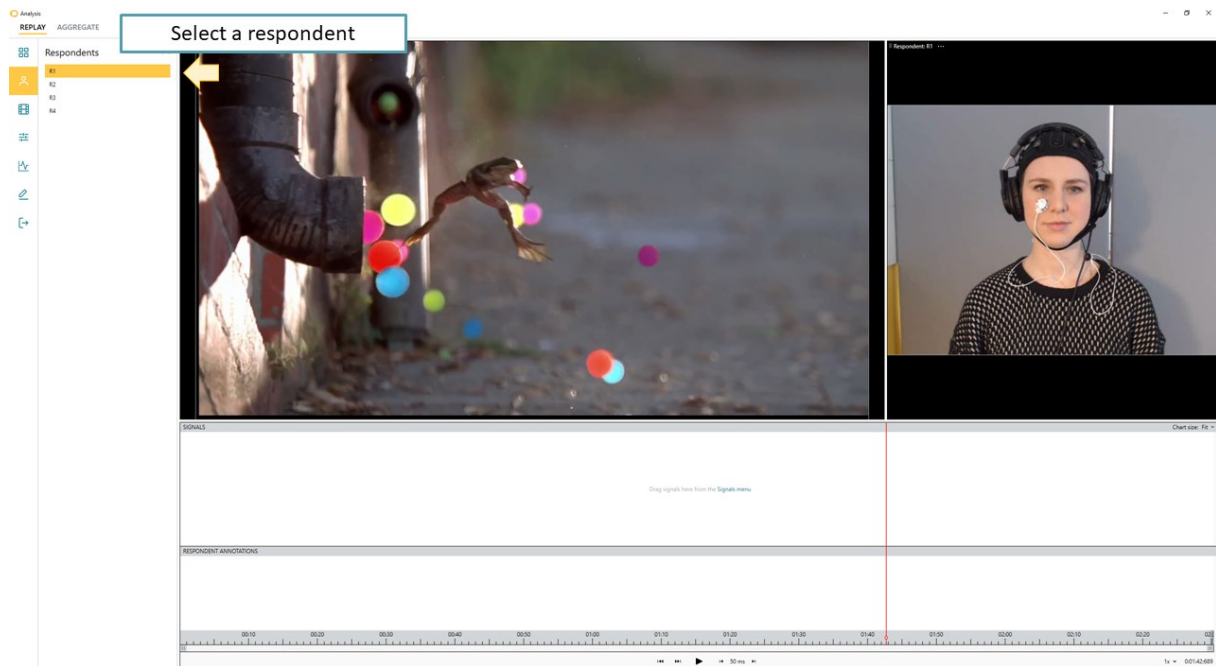
1. Grasp a panel by clicking in the top-left corner of a panel, and drop it in the panel you want to swap positions with.

Enlarge panels:

1. Grasp the border of a panel by clicking on it, and move it left or right (for vertical borders), or up and down (for horizontal borders).

Respondents

The 'Respondents' menu then allows you to navigate from respondent to respondent.



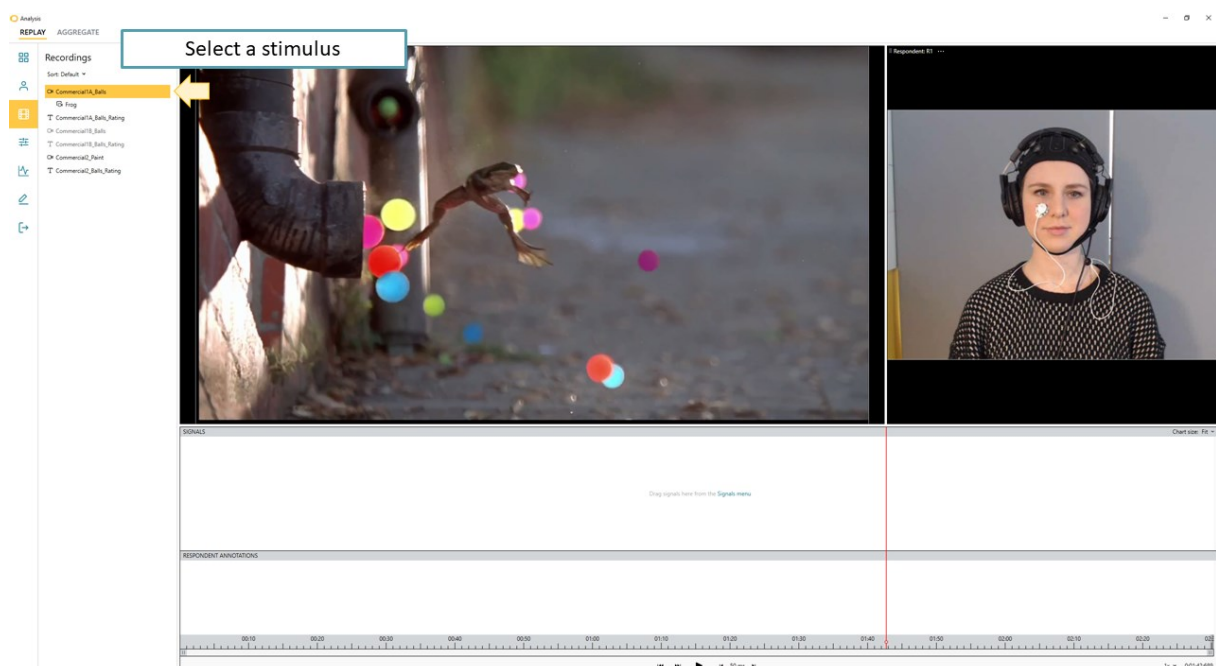
How to navigate from respondent to respondent

Select a respondent:

1. Navigate to the Respondents menu
2. Click on the name of a respondent in the list

Recordings

The 'Recordings' menu allows you to navigate from stimulus to stimulus.

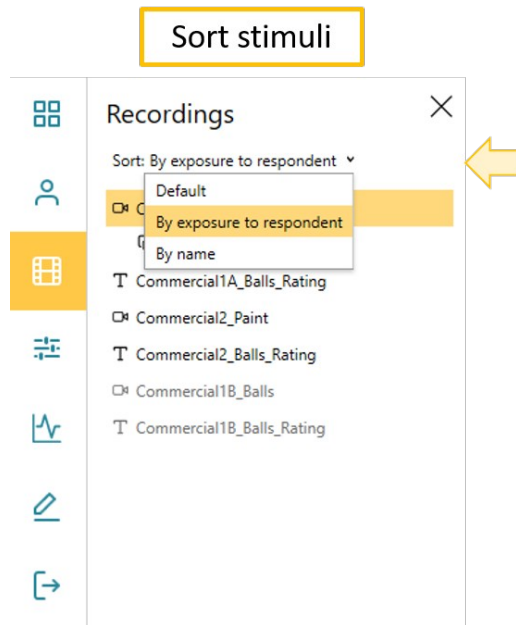


How to navigate from stimulus to stimulus

Select a stimulus:

1. Navigate to the Recordings menu
2. Click on a stimulus in the list

Note also that you can choose to sort the stimuli 'by name', or 'by exposure to respondent'. By default, the stimuli are sorted in the same way as they are ordered in the Stimulus bar. However, the option 'by name' and 'by exposure to respondent' allow you to sort the stimuli alphabetically or according to their order of occurrence during the data collection.



Sort the stimuli:

1. Navigate to the Recordings menu
2. Expand the sort option
3. Choose one of the options: 'Default', 'By exposure to respondent', or 'By name'

Visualization

The 'Visualization' menu is where you generate scan-paths (see also: [Scan-paths](#)). This is done by ticking the option 'Gaze visualization'. These scan-path can be static (i.e., summarizing the eye tracking data as an image) or dynamic (i.e., replaying the eye tracking data as a video), and overlay the stimulus in the Stimulus panel.

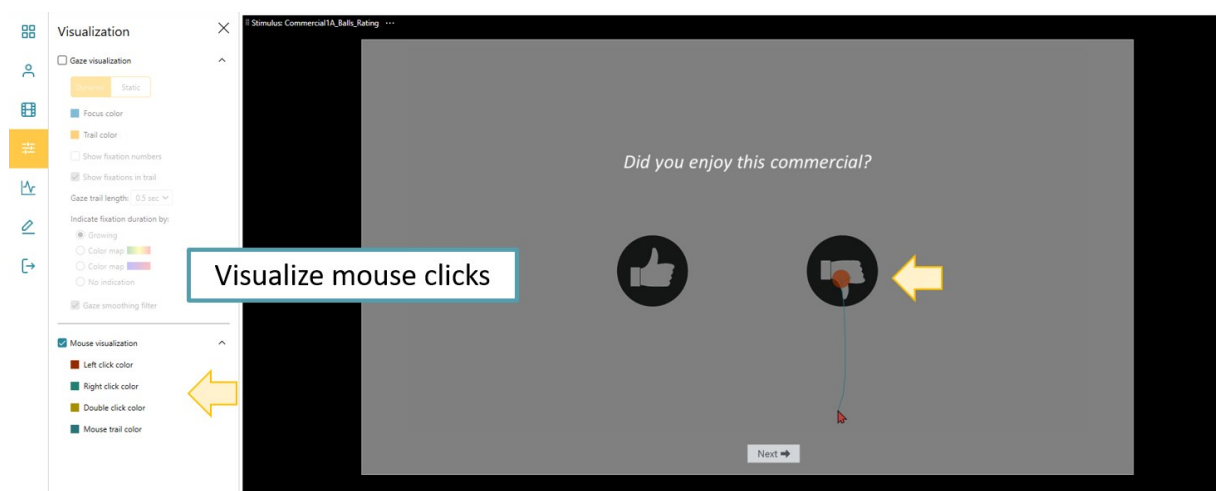


How to generate a scan-path

Make a scan-path:

1. Navigate to the Visualization menu
2. Tick gaze visualization
3. Consider the options (see also: [Scan-paths](#))

It is further also possible to display mouse clicks, by ticking the option 'Mouse visualization'.



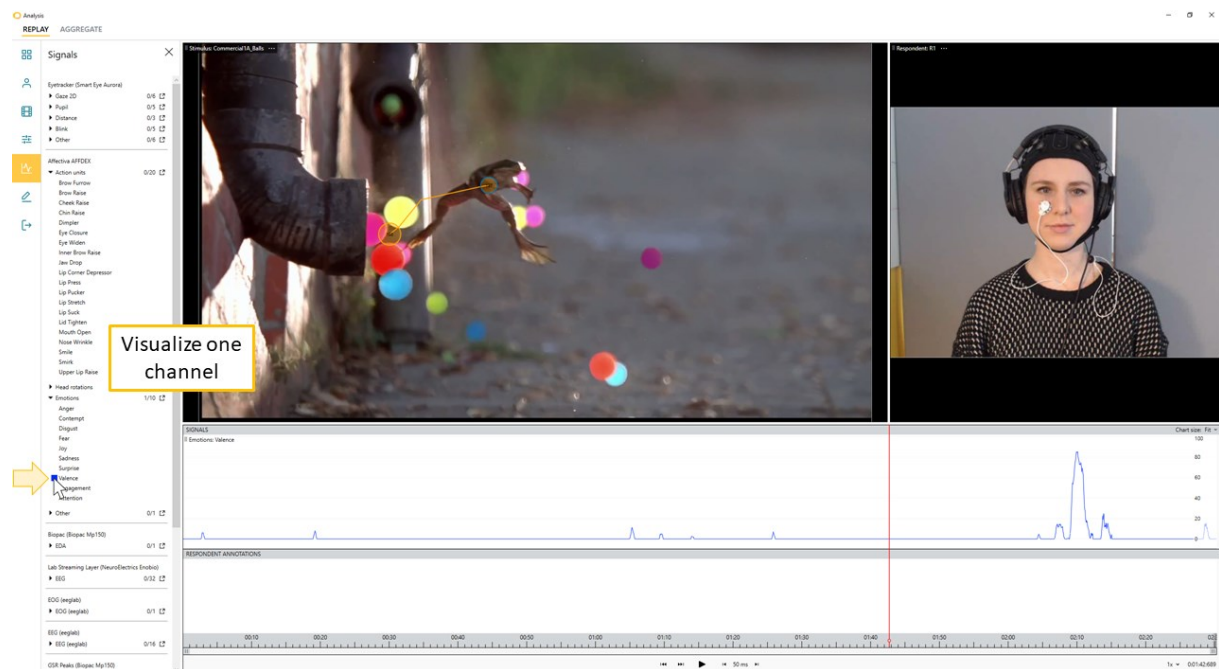
How to visualize mouse clicks

Visualize mouse clicks:

1. Navigate to the Visualization menu
2. Tick the option 'Mouse visualization'

Signals

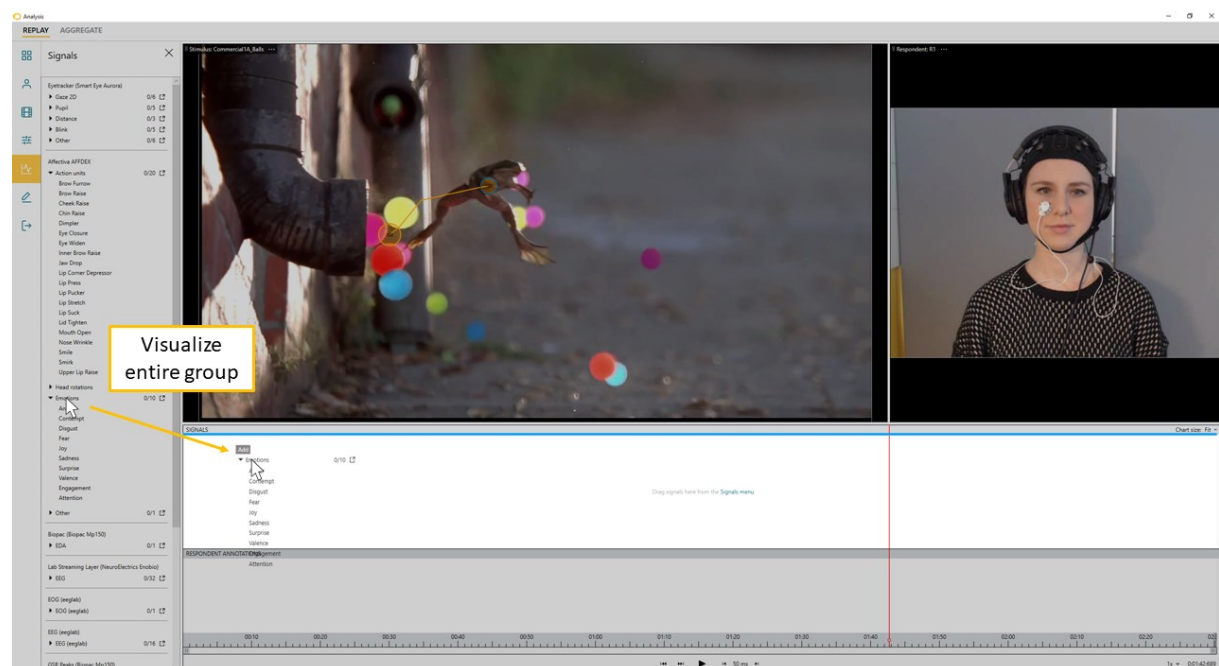
The 'Signals' menu allows you to select the signals to display as graph. These signals are organized in groups, and it is possible to add *individual* data-channels to the graph or immediately the entire *group* of data-channels.



How to add one data-channel to the Signals panel

Add one data-channel:

1. Expand a data section in the Signals list
2. Click on a specific channel. Alternatively, drag the channel and drop it in the Signals panel

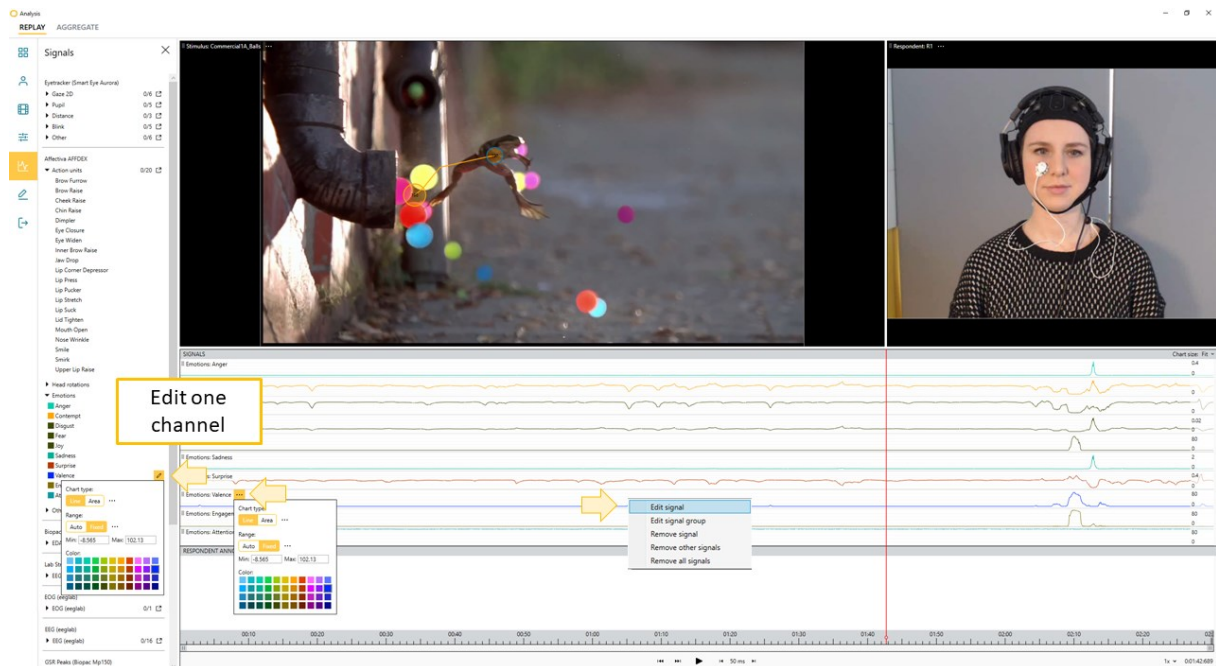


How to add a group of data-channels to the Signals panel

Add a group of data-channels:

1. Click on a data section in the Signals list, and drag the entire group of channels into the Signals panel

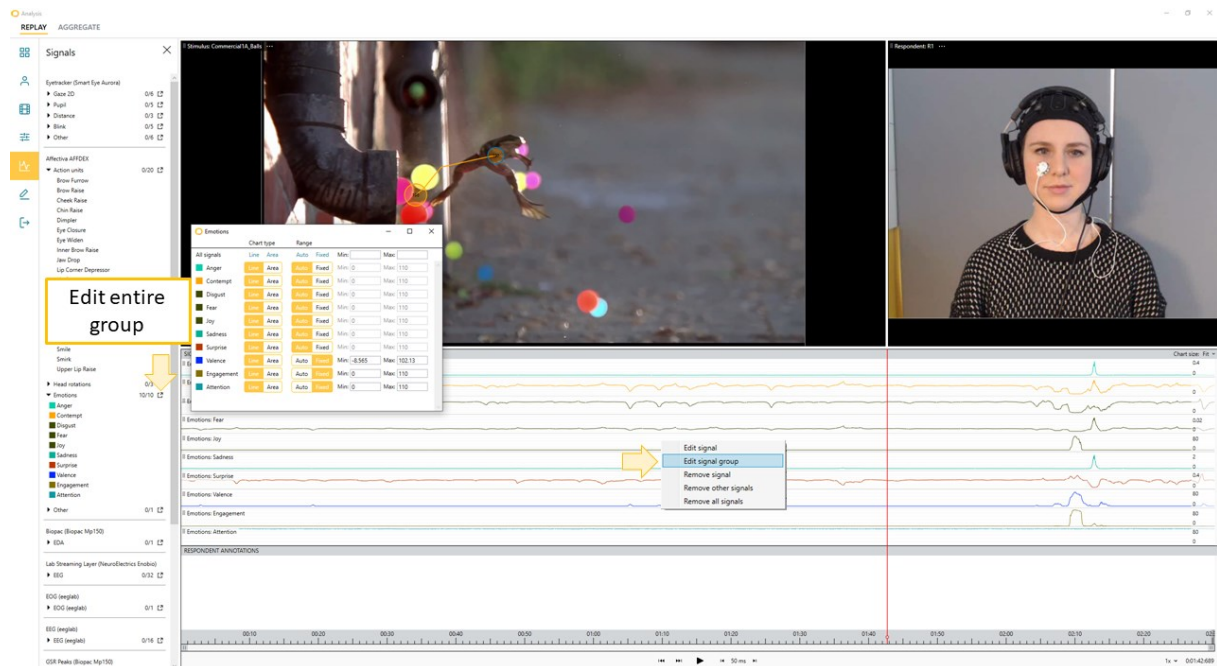
Also the editing of the graphs can be done per *individual* data-channel or immediately for the entire *group* of data-channels.



How to edit a graph

Edit a graph:

1. Click on the edit button for a specific channel, or on the '...' icon. Alternatively, right-click in the graph and choose Edit signal.
2. Review the options for editing: Chart type, Range, Color.



How to edit the graphs of a group of data-channels

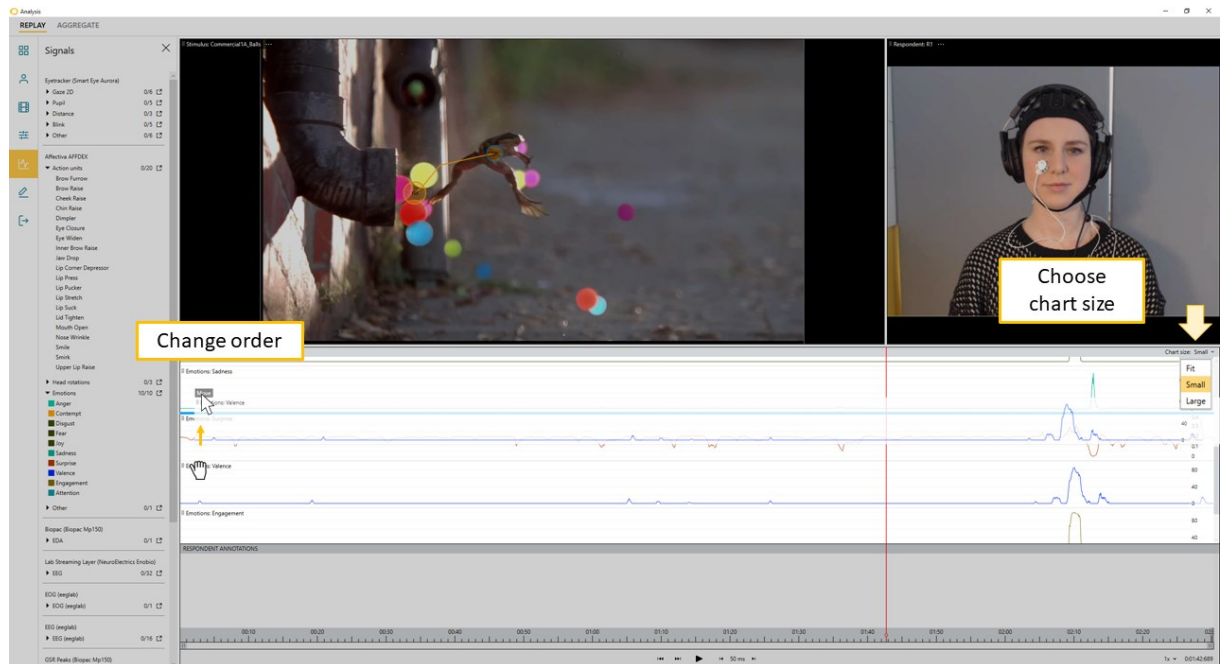
Edit the graphs of a group of data-channels:

1. Click on the edit button for an entire data section. Alternatively, right-click in the graph and choose Edit signal group.
2. Review the options for editing: Chart type, Range, Color.

Options:

Chart type	Choose between line or area plot
Range	Choose to auto-scale the graph, or set a fixed range (min-max)
Color	Choose a color

To change the size of the graphs you could play around with the 'fit', 'small' and 'large' visualization options. You may also choose to change the order of the graphs.



How to change the order of the graphs, and their size

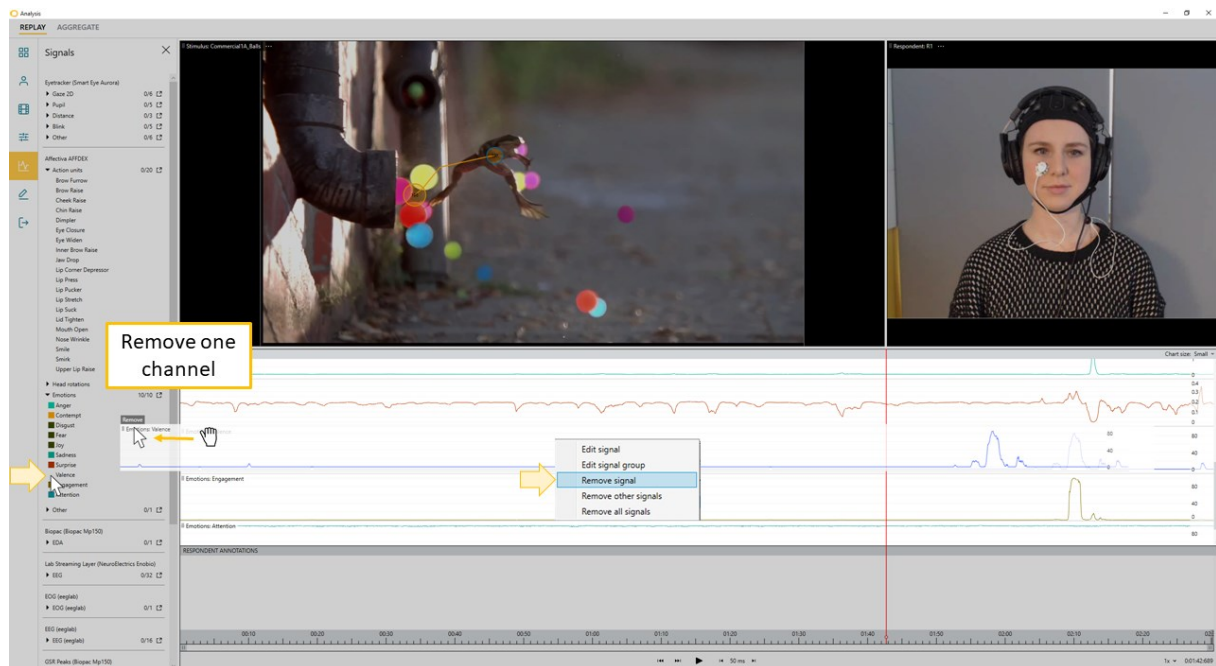
Change the order

1. Grasp a graph by clicking in the top-left corner of the graph, and move it up or down.

Change the chart size:

1. Expand the Chart size drop-down menu
2. Select 'Fit', 'Small' or 'Large'

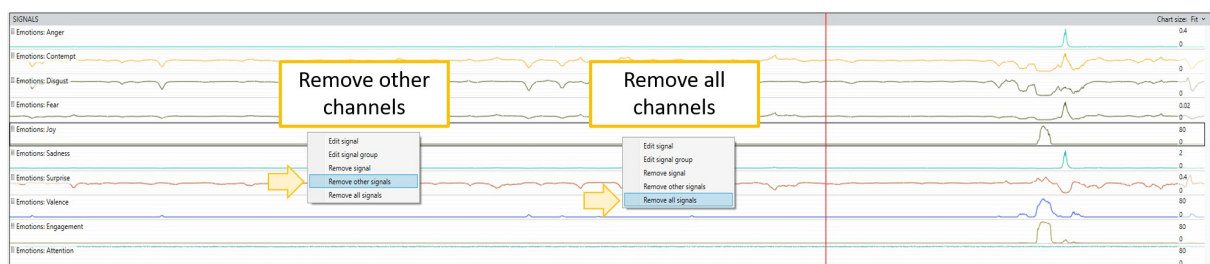
And finally, if you want to remove data-channels from the Signals panel, this is done in one of the following ways.



How to remove one data-channel from the Signals panel

Remove one data-channel:

1. Expand a data section in the Signals list
2. Click on a specific channel (the colored square in front of the channel disappears). Alternatively, grasp the graph by clicking on the name of the data-channel, and drop it outside of the Signals panel. Or, right-click in the graph, and choose Remove signal



How to remove all data channels except one, or all data channels

Remove all data channels except one:

1. Right-click in the graph that you wish to keep, and choose Remove other signals

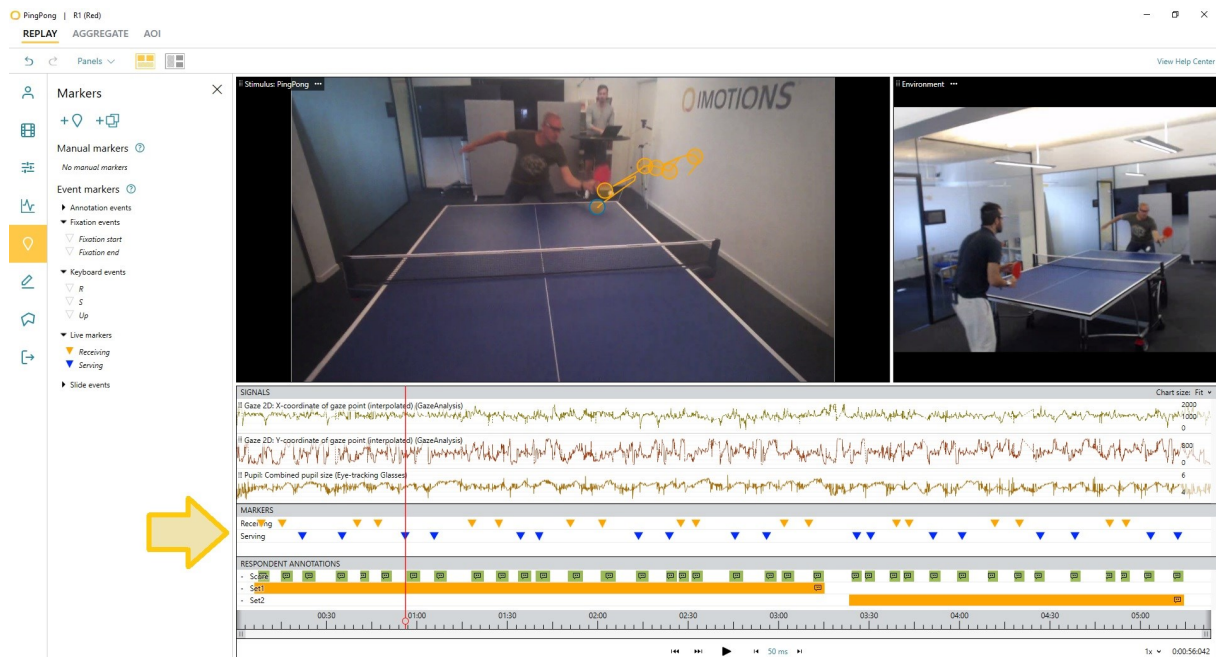
Remove all data channels:

1. Right-click anywhere in the Signals panel, and choose Remove all signals

Markers

Markers mark a point in time. There are two kinds of markers:

- Event markers were recorded during the data collection (such as **Live Markers**, keypresses, mouse clicks, **API events**, etc.) or generated from recorded sensor data (such as fixations from eye tracking data)
- Manual markers can be generated during the analysis



The Marker panel in iMotions

Event markers cannot be changed. It is possible to create a duplicate of an Event marker, which then can be edited the same way as a Manual marker.

Manual markers can be added to any study by clicking the **+📍** icon.

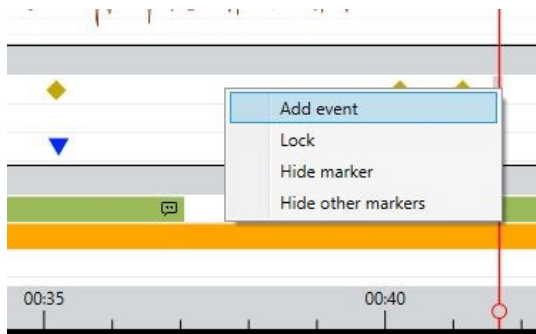
As an option, it is possible to arrange the Manual markers in groups. New groups can be created by clicking the **+📁** icon.



Manual Markers in a study

Setting new marker events is easiest through a keypress. You can choose a key when creating the new Manual marker.

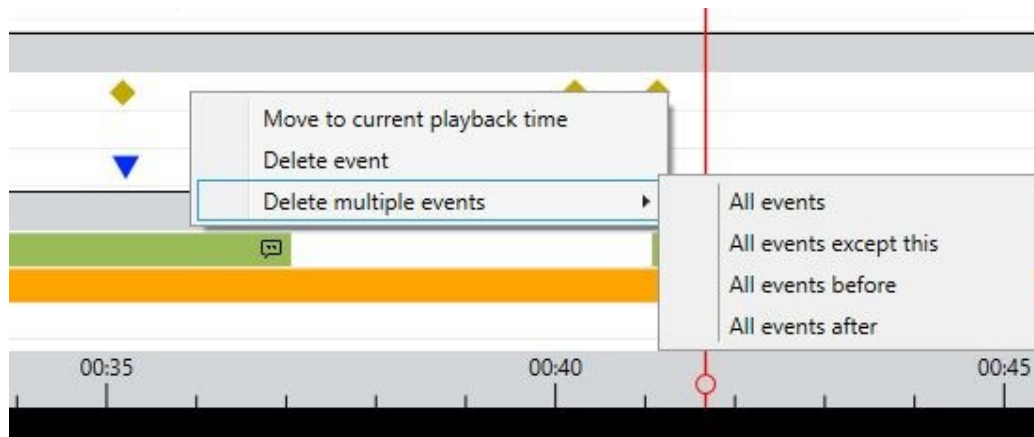
Alternatively, you can add new marker events by right clicking on the cursor.



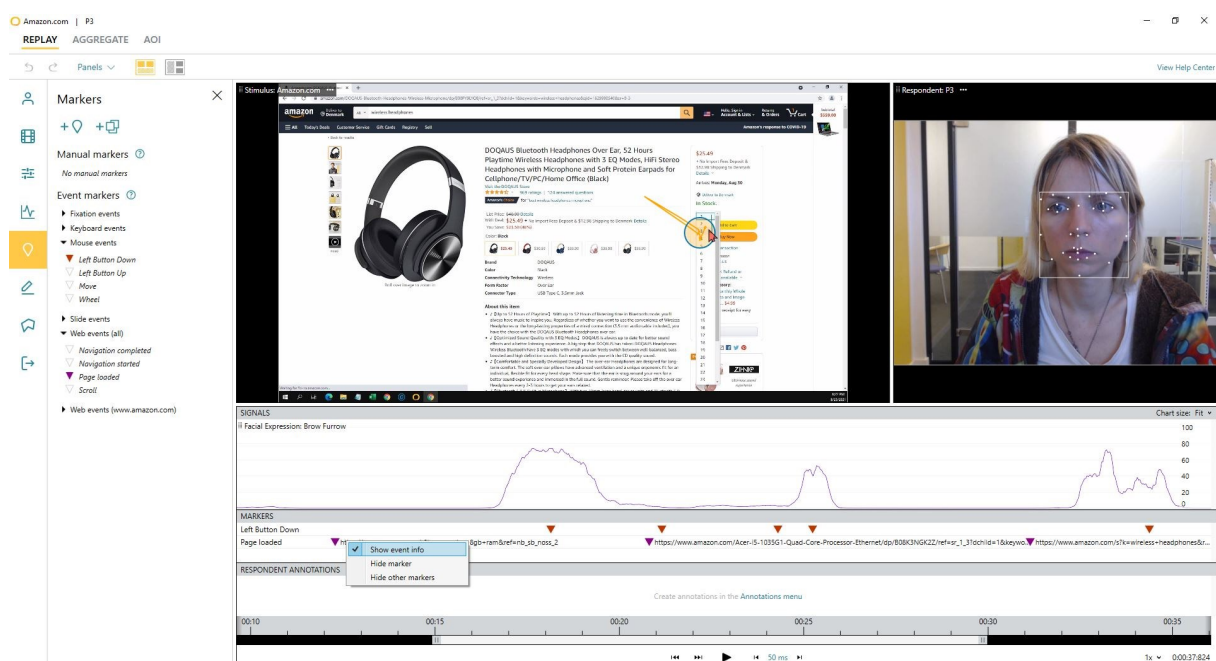
If you wish to edit a marker event, you can

- Delete the current marker
- Delete other markers than the selected one
- Move it to the current cursor time

by right clicking on the event

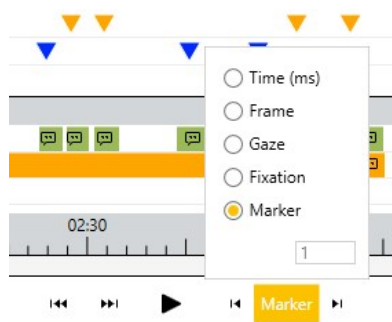


For some event types, it is possible to review additional data that was recorded with the event, such as the URL of a website.



Website events in the Events panel

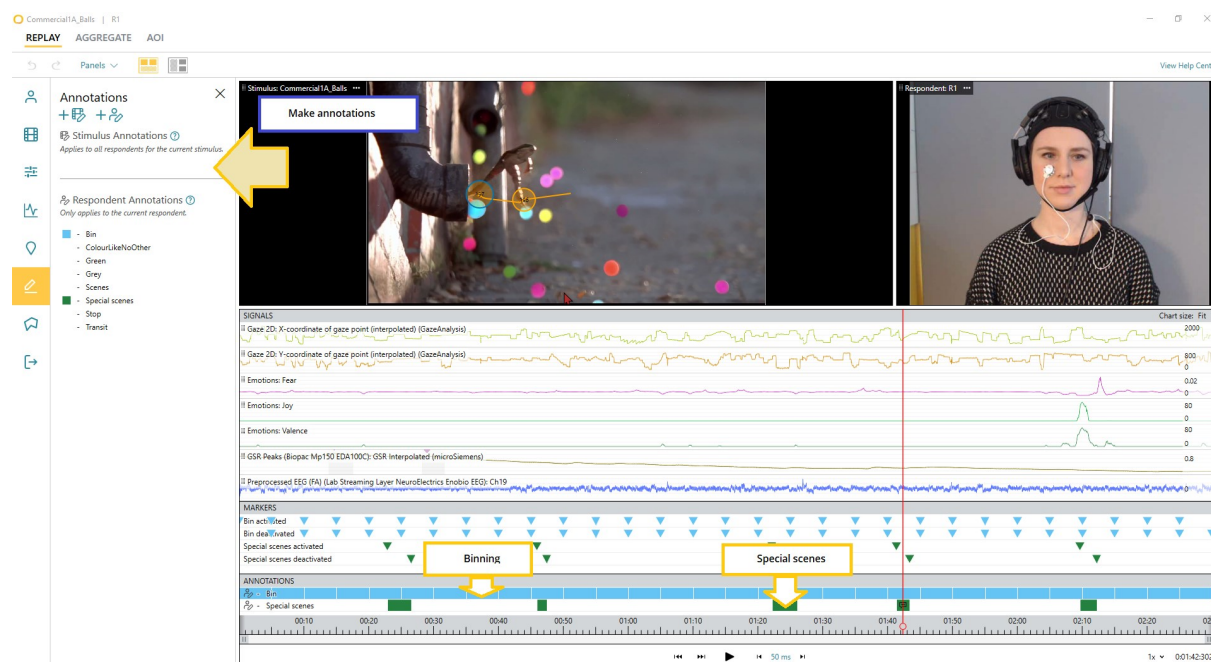
For all markers, you can step through the selected ones using the arrow keys or button controls.



The marker functionality is available in iMotions 9.1.12 and forward.

Annotations

The 'Annotations' allows you to annotate the recording. The annotations tool is a versatile tool that could be used for human behavior coding, for 'epoching', or for 'binning' the data.



How to add annotations

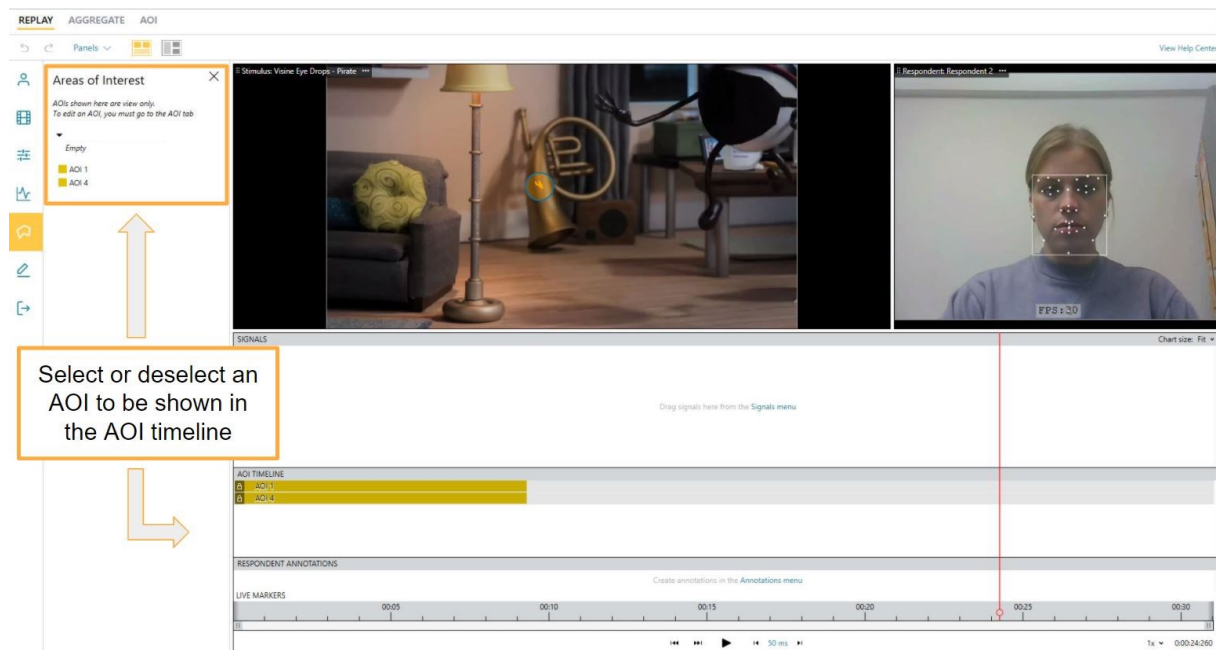
Add annotations:

1. Navigate to the Annotations menu
2. Define an annotation by clicking 'Add annotations', and start annotating. You can choose between Respondent annotations, which will only mark data for the current Respondent, or Stimulus annotations, which will mark the data for all Respondents at the same time (given that the stimulus can be [aggregated](#)).

You can read more about Annotations in [this article](#).

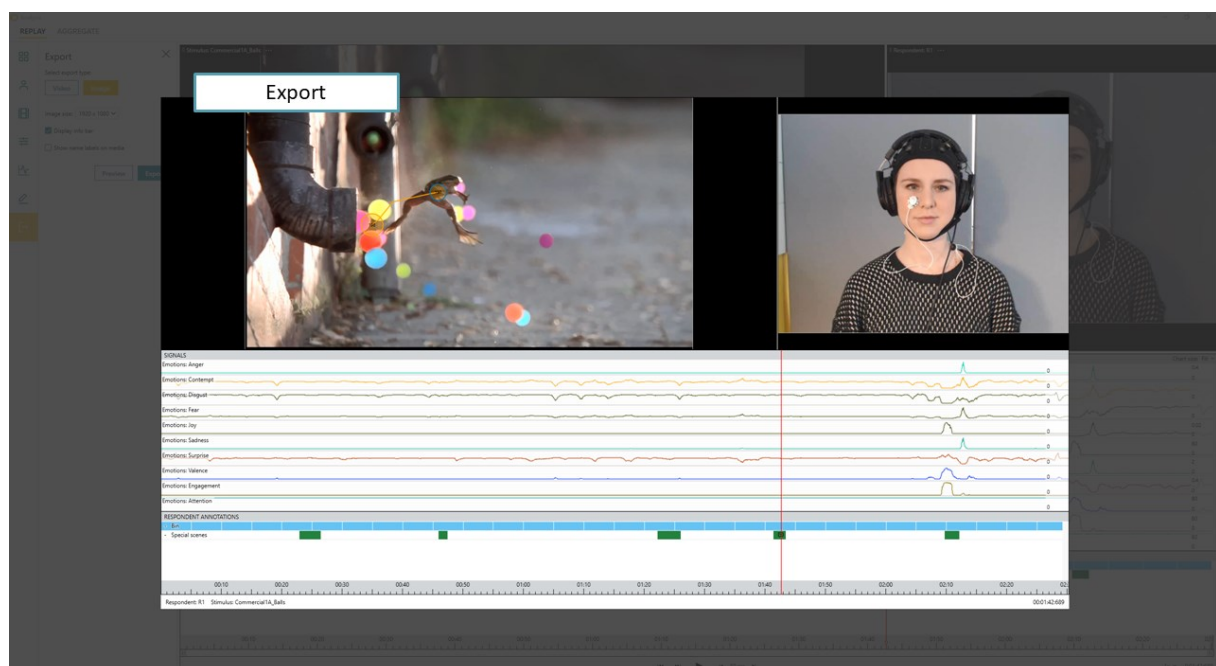
Areas of Interest

The 'Areas of Interest' section allows you to only view what AOIs have been made, and they can only be edited in the [AOI tab](#). Additionally, they can be selected or deselected in order to be shown on the AOI timeline.



Export

The 'Export' menu, finally, allows you to export images and videos from the Replay editor which you can share with colleagues or clients, or use them in a report or publication. For instance, you could choose to export images of the graphs, or videos of (dynamic) scan-paths.



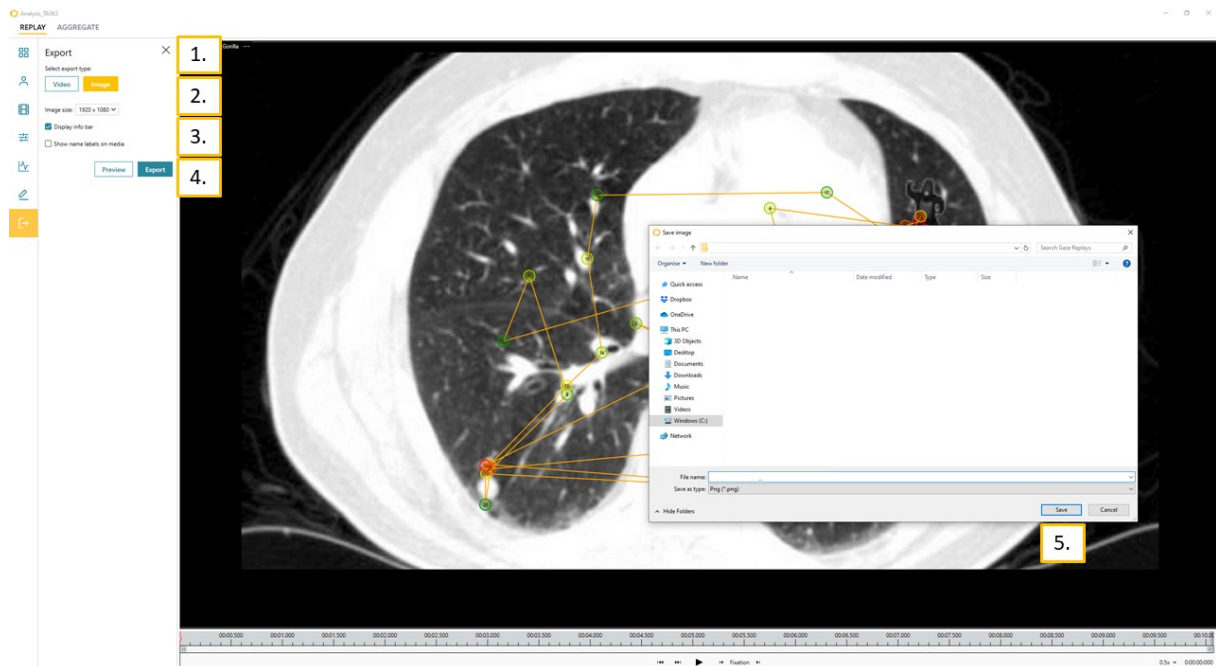
Preview of a graph export

Export an image (.png) or video (.wmv, .mp4):

1. Navigate to the Export menu
2. Select Video or Image

3. Click Export

- You have the option to either export the entire recording duration or just a selected time range
- You can choose how much you wish to zoom into the timeline
- The option to "Follow cursor" changes whether the timeline remains static and the red timeline cursor moves forward, or whether you wish the cursor to remain static and the timeline to move
- You can customize the quality and format of your exported video



To export the static scan-path as an image (yellow path):

1. Navigate to the Export menu
 2. Select 'Image' for exporting the static scan-path
 3. Review the export options
 4. Click 'Export'
 5. Save the image file on your computer
- The option to "Display info bar" adds a bottom bar with information about the current stimulus
 - The option to "Show name labels on media" includes the panel labels in the screenshot

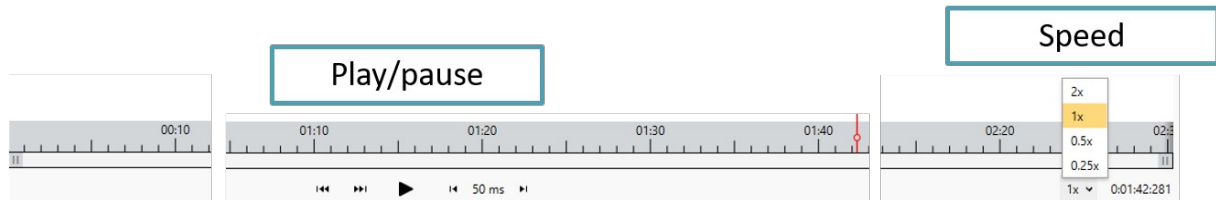
Good to know (features)

A few more features are good to be aware of when operating the Replay editor.

Play, pause, slow motion

Replaying the recordings is done by clicking the play button. If you click this button, the recordings in the different panels replay in synchrony; a red cursor follows the progression on the timeline.

It is further possible to replay the data in slow motion (0.25x, 0.5x), or in fast motion (2x, 4x).



How to replay the recordings at regular speed, or in fast/slow motion

Play/pause the recordings:

1. Click play/pause

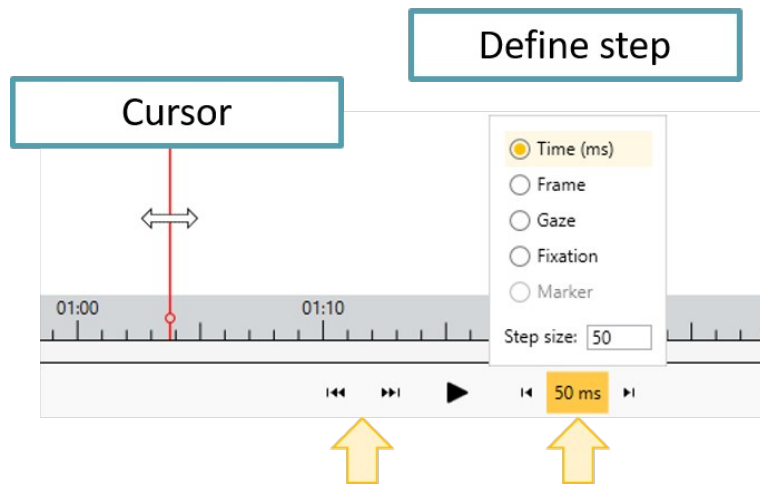
Play in fast/slow motion:

1. Expand the speed options
2. Choose a desired speed (0.25x, 0.5, 1x, 2x, 4x).

Navigate with precision

Apart from replaying the recordings, you may also want to step through the data by moving the cursor. You could jump to the start or end of the recording by using the 'Go to start' and 'Go to end' buttons, or you could place the cursor in any location by clicking on the timeline.

If you want to navigate through your data in a more controlled fashion, you could also set a desired step size, and then, either step through the data by using the 'Step back' and 'Step forward' buttons, or by using the arrow keys on the keyboard. It is possible to step by time (in milliseconds), frame-per-frame, per gaze point, per fixation or, if you have defined Live markers (see also: [Live markers](#)), from marker to marker.



How to navigate through the recordings with precision

Step through the data:

1. Click anywhere on the timeline, or grasp the cursor and drag it over the timeline

Step through the data more precisely:

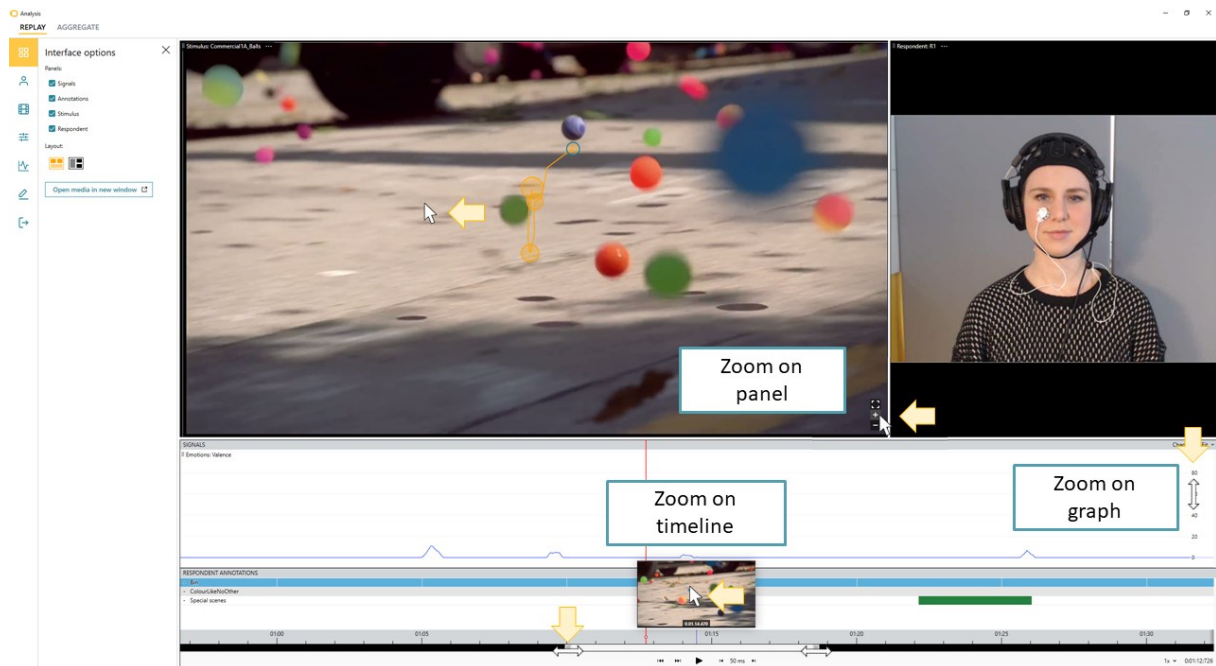
1. Expand the step options
2. Choose a desired step: Time (ms), Frame, Gaze, Fixation, or Marker. If you choose Time (ms), you should also define the time in milliseconds.
3. Now, click the 'Step back' and 'Step forward' buttons to navigate through the recording. Alternatively, use the arrow keys on the keyboard.

Jump directly to the start or end of the recording:

1. Click on the 'Go to start' or 'Go to end' button.

Zoom

It is further possible to zoom on the timeline, to zoom in on a panel ('Stimulus', 'Respondent', 'Environment'), or to zoom in on a graph ('Signals').



How to zoom

Zoom on the timeline:

1. Place your mouse anywhere in the area that covers the timeline, Annotations or Signals panel.
2. Scroll up or down. Alternatively, drag the borders of the zoom bar underneath the timeline.

Zoom on a recording ('Stimulus', 'Respondent', 'Environment'):

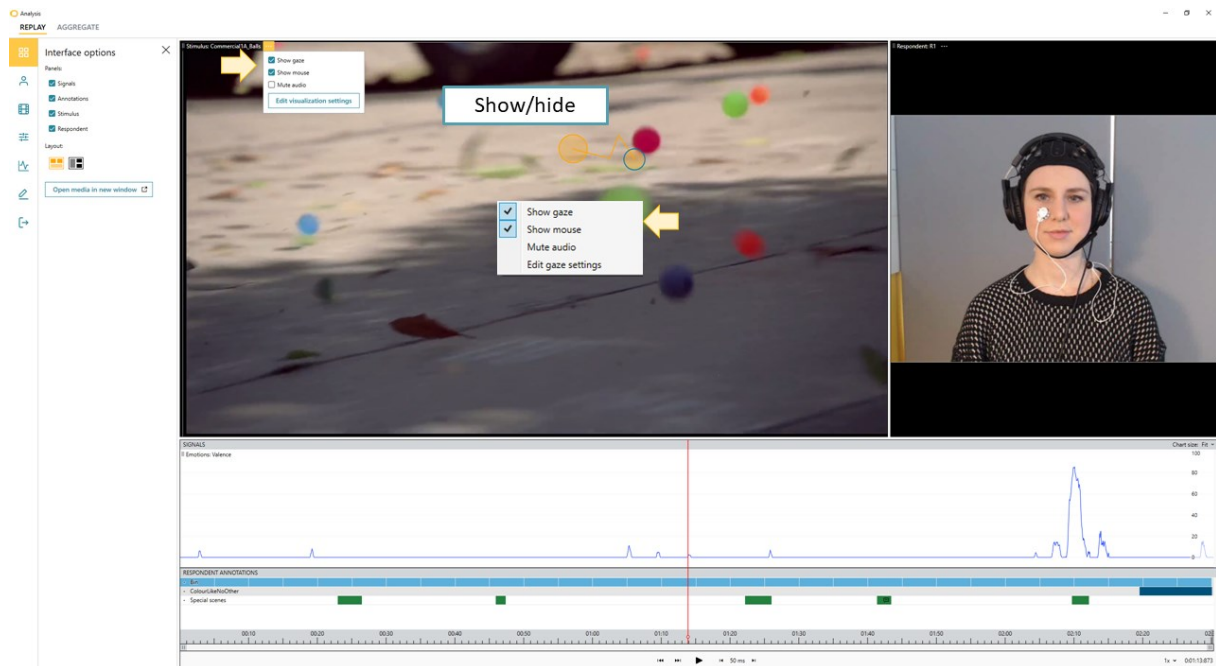
1. Place your mouse over the panel of interest
2. Scroll up or down. Alternatively, click the '+' or '-' buttons in the bottom-right corner of the panel.

Zoom on a graph ('Signals'):

1. Click on the scale of a graph.
2. Hold the mouse click, and move up or down. Note that you could also edit the scales of the graphs in a more controlled way (see below: [Signals](#))

Show/hide

If you have recorded eye tracking or mouse data, but you want to hide the visualization of it (as it overlays the Stimulus), you can right-click in the Stimulus panel to hide the overlay. You could also navigate to the 'Visualization' panel to disable the gaze and/or mouse visualizations from there (see below: [Visualization](#)).



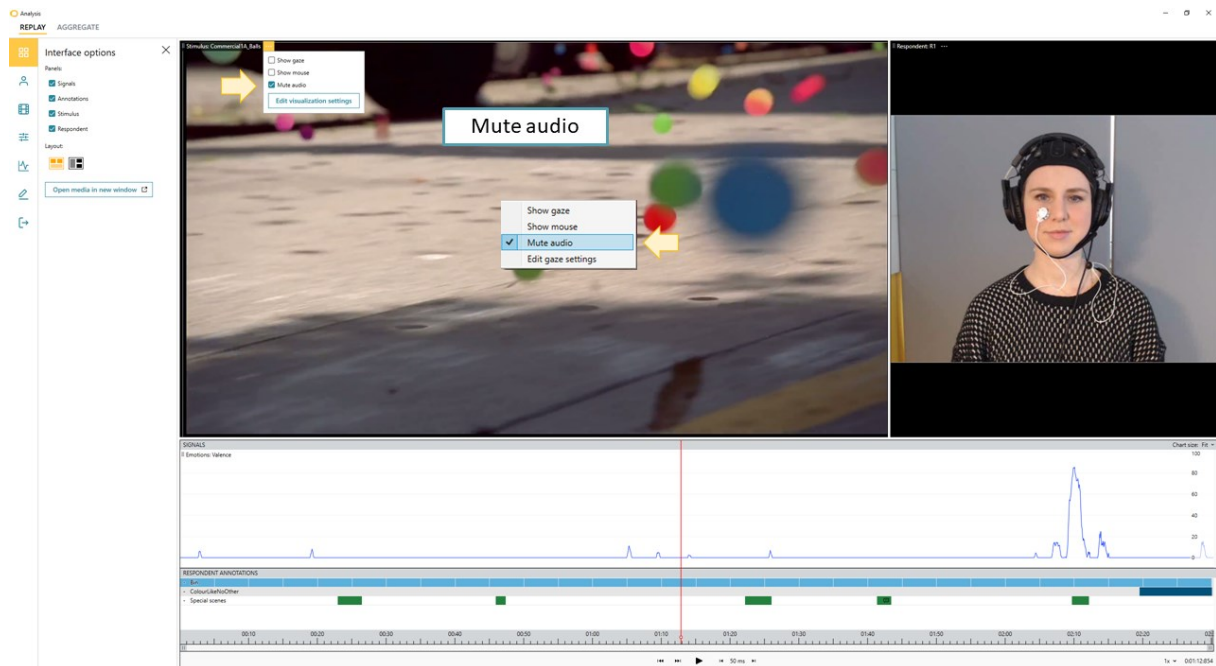
How to show/hide gaze and mouse visualizations

To show/hide gaze or mouse visualizations:

1. Right-click in the 'Stimulus' panel. Alternatively, click on the '...' icon in the top-left corner of the panel.
2. (Un)tick Show gaze and/or Show mouse, or click Edit visualization settings to navigate to the 'Visualization' panel.

Mute

If your recordings have audio, and you want to mute the sound, you can right-click in the panel to mute.



How to mute the audio

Mute the audio from a recording ('Stimulus', 'Respondent', 'Environment'):

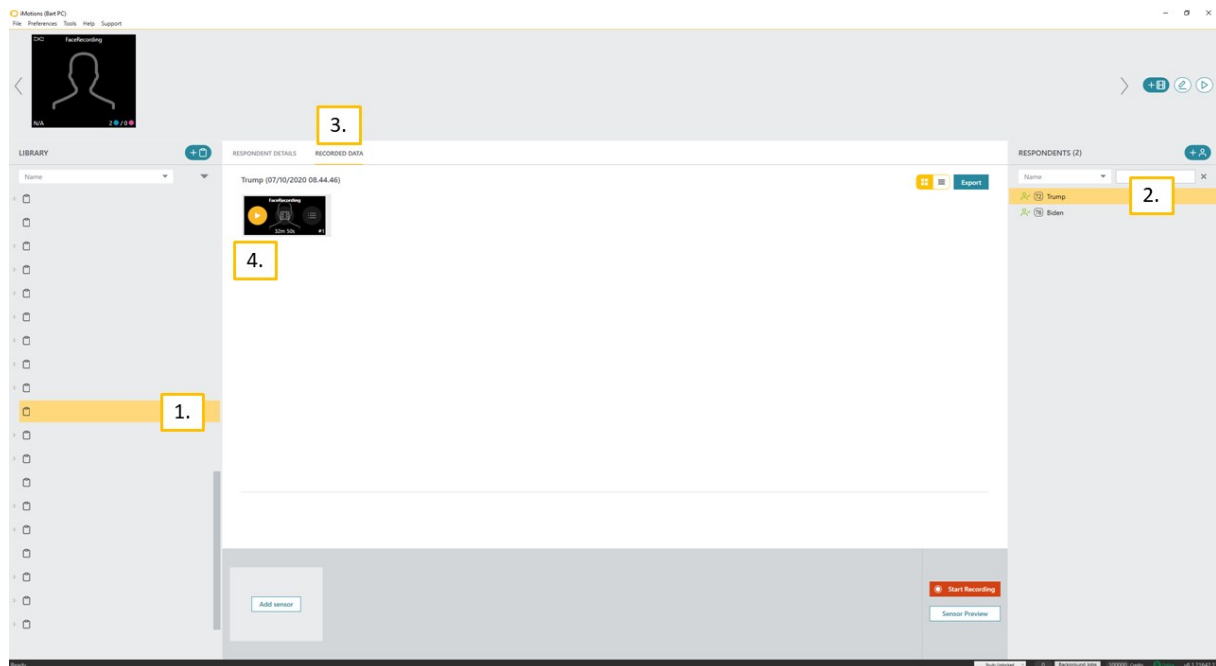
1. Right-click in the panel to mute. Alternatively, click on the '...' icon in the top-left corner of the panel.
2. Click Mute audio.

Define a Facebox

If you need to **post process the respondent video** to extract facial expressions, you can define a facebox around the face of interest. By defining a facebox, you constrain the area where the algorithm looks for the presence of a face. Hence, rather than to look for a face in the entire frame of the video, the algorithm will now only analyze the face(s) that is(are) present within the area of the facebox.

In other words, defining a facebox is a way to make sure that, at all times the correct face is being analyzed - or equivalently, to avoid the algorithm detaching from the face of interest and accidentally start analyzing the wrong face.

Defining a facebox is something you do *before* post-processing the face recordings, from within the Replay editor.



Access the Replay editor from the 'Study level' (yellow path):

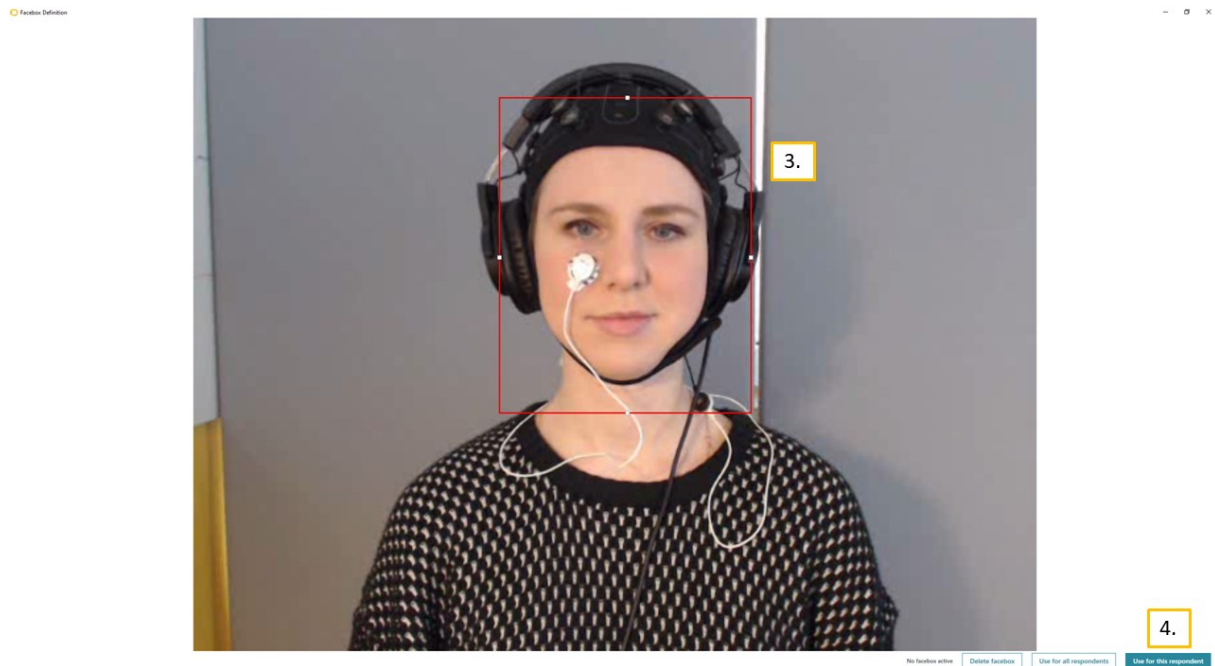
1. Select the study in the Library
2. Select a respondent from the Respondents list
3. Navigate to the 'Recorded data' page in the Respondent menu
4. Click the 'Play' button for a given stimulus



Define a facebox (yellow path):

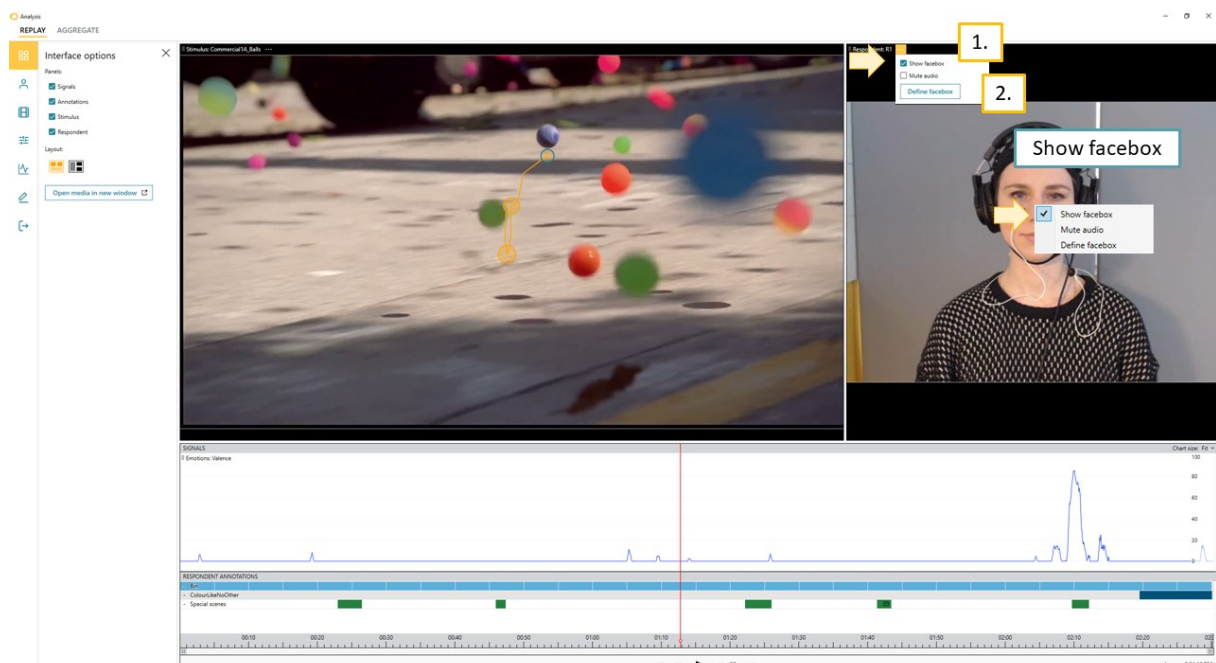
1. Right-click in the 'Respondent' panel. Alternatively, click on the '...' icon in the top-left corner of the panel
2. Click Define facebox
3. Position the facebox
 - a. Place the (red) square around the face of interest (there could be more than one, depending on the stimulus)
4. Choose Use for this respondent or Use for all respondents. It is also

possible to delete a previously defined facebox.



Show/hide the facebox

Finally, if you have have applied facial expression analysis, but you want to hide the facebox (that overlays the recording of the Respondent), you can right-click the the Respondent panel to hide the facebox.



How to show/hide a facebox

Show/hide the facebox:

1. Right-click in the 'Respondent' panel. Alternatively, click on the '...' icon in

- the top-left corner of the panel
2. (Un)tick Show facebox

Getting started video

For an introduction to the Replay, you can watch this webinar by our Customer Success Manager Amie Larum (recorded in iMotions 8.1, February 2020)

If you want to play around with the Replay editor yourself, please consider downloading the [Sony Bravia \(Testing of video commercials\)](#), [Inattentional Blindness \(Psychology experiment\)](#), [In-store VR \(Shelf-testing in VR environment\)](#), or [Strooptest \(Psychology experiment\) demo-studies](#).

